



Lesson 5.1: History of Computers

https://codehs.com/lms/assignment/93877803/lesson_plan

Description	<i>When was the first computer made? What did it look like, and what was it used for?</i> In this lesson, students will explore the creation and evolution of computing machines that now permeate our day-to-day life.
Objective	Students will be able to: • Identify important historical events in the development of modern computers • Explore individual's contributions to the development of the computer and discuss who gets to be included in the <i>computer innovators</i> group
Activities	2.1.1 Video: History of Computers 2.1.2 Check for Understanding: Video Quiz 2.1.3 Notes: Mission: Who invented the computer? 2.1.4 Free Response: Evidence Collection 2.1.5 Connection: Exhibit A: Charles Babbage 2.1.6 Connection: Exhibit B: Ada Lovelace 2.1.7 Connection: Exhibit C: Alan Turing 2.1.8 Connection: Exhibit D: Mauchly and Eckert 2.1.9 Connection: Exhibit E: ENIAC Programmers 2.1.10 Connection: Exhibit F: Grace Hopper 2.1.11 Connection: Exhibit G: Mark Dean 2.1.12 Connection: Bonus Exhibit: Computer Inventors 2.1.13 Free Response: Culminating Activity
Prior Knowledge	No prior knowledge needed.
Planning Notes	• Update! This lesson was updated on October 7, 2020. You can find the original material in the Supplemental Module titled "Original Material: What is Computing?" • Review the slides and the exercises in the lesson. • This lesson is an exploratory lesson where students investigate the question <i>Who invented the computer</i>. Students will review a series of 8 exhibits, each of which describe a specific person or group of people. While the number of exhibits may appear overwhelming, each exhibit should not take more than 5-7 minutes. You can structure this activity in multiple ways: ◦ Give students a specific amount of time per exhibit.

- Give students a specific amount of time for the entire investigation, enabling students to work the exhibits at their own pace.
 - Discuss each exhibit as a class.
- There is no right or wrong answer to how students approach answering the question *who invented the computer?* Because the computer was developed by all of these amazing innovators, students can easily defend one person or they could argue that there is no one person who invented the computer.
- In the *Bonus Exhibit* activity, students explore who is included and who is excluded from the typical group of *computer inventors*. The purpose of this activity is to challenge the belief that all of the main computer inventors were white men - women and people of color made significant contributions as well! If time permits, discuss student responses as a class. Because the people excluded are females and people of color, be prepared to lead a potentially sensitive discussion. Set discussion expectations to ensure students listen to and respect each other's opinions and feel safe sharing their thoughts.
- Encourage students to open the *Evidence Collection* activity in a separate tab so they can complete it as they go through each exhibit.
- There is a handout that accompanies this lesson. It can be used as an in-class activity or a homework assignment. Determine how and if this handout will be used and make the appropriate number of printouts prior to the class period.

Standards Addressed

CSTA 2 Standards

Name	Description
2-IC-20	Compare tradeoffs associated with computing technologies that affect people's everyday activities and career options.

PRAXIS 5652 Standards

Name	Description
I.A.1	Understand computing as a way of expressing creativity, solving problems, enabling communication, and fostering innovation in a variety of fields and careers
I.A.1.b	recognize the benefits of using computers to solve problems
I.A.1.d	provide examples of how computers foster innovation
I.A.3.a	analyze computing innovations in terms of their social,

	<table> <tr> <td data-bbox="472 73 962 170"></td><td data-bbox="962 73 1479 170">economic, and cultural impacts, both beneficial and harmful</td></tr> <tr> <td data-bbox="472 170 962 327">I.A.3.b</td><td data-bbox="962 170 1479 327">identify trade-offs between beneficial and harmful effects of computer innovations</td></tr> </table>		economic, and cultural impacts, both beneficial and harmful	I.A.3.b	identify trade-offs between beneficial and harmful effects of computer innovations
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Teaching and Learning Strategies	Lesson Opener: - Have students brainstorm and write down answers to the discussion questions listed below. Students can work individually or in groups/pairs. Have them share their responses. [5 mins] Activities: - Watch the lesson video and take the corresponding quiz. This quiz is a quick check for understanding. [10-15 mins] - Complete the <i>Mission: Who invented the computer</i> Activity [2-5 mins] - Review the activity directions with students. - Encourage students to open the <i>Evidence Collection</i> activity in a new tab so they can write down their answers to the guiding questions as they go. - Give students time to explore Exhibits A-G [50-60 mins] - Each exhibit should take about 5-7 minutes. You can have students go through each at their own pace or give them a specific amount of time to investigate all exhibits. - If time permits, discuss the Bonus Exhibit with students. Because this touches on race and gender, be prepared to lead a sensitive discussion. - Complete the <i>Culminating Activity</i> [10-20 mins] - Give students time to complete the activity independently. - If time permits, discuss students' conclusions as a class. Encourage students to challenge and build on each other's responses. Lesson Closer: - Have students reflect and discuss their responses to the end of class discussion questions. [5 mins]				
Discussion Questions	Beginning of class: - What is the oldest computer or piece of technology in your house? <i>Sample Response: My graphing calculator is pretty old.</i> - What does the term "computer" mean to you? <i>Answers will vary.</i> - Display an image of the ENIAC without telling students what it is. What is this a picture of? <i>Answers will vary.</i> End of class:				

	• How have computers changed since the first computing machines? ◦ <i>Computers take up less space and run faster. They are able to “read code” instead of using punch cards to program a computer.</i> • Think of a task that you recently used a computer for. If you didn’t have a computer to help you complete this task, how would you have done it differently? ◦ <i>Sample Response: If I didn’t have a Word Processor, I would have to write essays and schoolwork by hand.</i> • What was something interesting that you learned about the history of computers? ◦ <i>Answers will vary.</i>
Resources/Handouts	Computer History Timeline (teacher version) Computer History Timeline (student version)

Vocabulary

Term	Definition
Computer	A person or device that makes calculations, stores data, and executes instructions according to a program.

Modification: Advanced	Modification: Students Needing Additional Support	Modification: English Language Learners
• Encourage students to explore who else should be included in the list of computer inventors	• Print out video slides for students to reference • Strategically pair students during the exhibit investigations • Differentiate the articles so that students can access all of the material	• Print out video slides for students to reference • Strategically pair students during the exhibit investigations • Provide students with a dictionary of their primary language